

Mini Project Report: Comparative Analysis of Abalone and Crab Age Prediction Datasets

Course: CSE 303

Team: [Your Team Name]

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1. Introduction

This project presents a comprehensive comparative analysis of two marine creature datasets: the Abalone dataset from UCI Machine Learning Repository and the Crab Age Prediction dataset from Kaggle. The primary objective is to develop and compare machine learning models for age prediction in these marine species, utilizing both regression and classification approaches.

The significance of this study lies in understanding biological age indicators across different marine species and evaluating the effectiveness of various machine learning algorithms in predicting age based on physical characteristics. Age prediction in marine biology is crucial for population studies, conservation efforts, and understanding ecosystem dynamics.

Our analysis employs two fundamental machine learning approaches: Linear Regression for continuous age prediction and Logistic Regression for binary age classification (young vs old). This dual approach provides insights into both precise age estimation and categorical age grouping, which are valuable for different research applications.

2. Dataset Description

2.1 Abalone Dataset

The Abalone dataset contains information about abalone shells with the following characteristics:

- **Total Samples:** 4,177 records
- **Features:** 8 attributes plus target variable
- **Target Variable:** Rings (used to determine age)
- **Data Types:** Mixed (categorical and numerical)

Key Features:

- Sex: Categorical variable (M=Male, F=Female, I=Infant)
- Length: Longest shell measurement (continuous)
- Diameter: Shell diameter perpendicular to length (continuous)
- Height: Shell height with meat in shell (continuous)
- Whole weight: Whole abalone weight (continuous)
- Shucked weight: Weight of meat (continuous)
- Viscera weight: Gut weight after bleeding (continuous)
- Shell weight: Weight after being dried (continuous)
- Rings: Target variable for age prediction (+1.5 gives the age in years)

2.2 Crab Age Prediction Dataset

The Crab Age Prediction dataset focuses on crab age determination with:

- **Total Samples:** Varies based on dataset version
- **Features:** Multiple morphological measurements
- **Target Variable:** Age (direct age measurement)
- **Data Types:** Primarily numerical with some categorical variables

Key Features:

- Sex: Gender classification (male/female)
- Various morphological measurements (length, width, weight parameters)
- Age: Direct target variable representing crab age

Both datasets represent real-world biological data with inherent complexities including missing values, outliers, and categorical variables requiring preprocessing for machine learning applications.

3. Data Preprocessing

3.1 Missing Value Analysis

Our initial analysis revealed the data quality status for both datasets:

Abalone Dataset:

- Missing values detected using `df.isnull().sum()` method

- Percentage calculation performed using `df.isnull().mean() * 100`
- Visual analysis conducted using missingno library

Crab Dataset:

- Similar missing value detection methodology applied
- Comprehensive percentage analysis performed
- Visual representation created for better understanding

3.2 Missing Value Treatment

We implemented a systematic approach for handling missing values:

For Categorical Variables:

```
python  
  
mode_val = df_copy[column].value_counts()  
df_copy = df_copy.fillna({column: mode_val.index[0]})
```

For Numerical Variables:

```
python  
  
med_val = df_copy[column].median()  
df_copy[column] = df_copy[column].fillna(med_val)
```

3.3 Duplicate Handling

- Duplicate detection using `df.duplicated().sum()`
- Removal of duplicate entries to ensure data integrity
- Final dataset shapes verified after cleaning

3.4 Categorical Variable Conversion

Abalone Dataset:

- Sex variable mapped: M=0, F=1, I=2
- All categorical variables converted to numerical format for ML compatibility

Crab Dataset:

- Sex variable mapped: male=0, female=1
- Systematic conversion of remaining categorical variables using label encoding

3.5 Outlier Detection

Implemented IQR-based outlier detection:

```
python
def find_outliers_IQR(df):
    q1 = df.quantile(0.25)
    q3 = df.quantile(0.75)
    IQR = q3 - q1
    outliers = df[((df < (q1 - 1.5 * IQR)) | (df > (q3 + 1.5 * IQR)))]
    return outliers
```

This method identified outliers in key numerical features, providing insights into data distribution extremes.

4. Exploratory Data Analysis (EDA)

4.1 Correlation Analysis

Comprehensive correlation matrices were generated for both datasets to understand feature relationships:

Abalone Dataset Insights:

- Strong correlations observed between physical measurements (length, diameter, height)
- Weight measurements showed high intercorrelation
- Target variable (Rings) showed moderate correlations with size measurements

Crab Dataset Insights:

- Similar patterns in morphological measurements
- Age correlations with physical characteristics varied
- Sex differences in measurement correlations observed

4.2 Visualization Insights

Distribution Analysis:

- Pie charts revealed sex distribution patterns in both species
- Histograms with KDE showed age distribution characteristics
- Box plots identified quartile distributions and potential outliers

Relationship Analysis:

- Bar plots demonstrated average age differences by sex
- Count plots showed categorical distribution patterns
- Scatter plots revealed linear and non-linear relationships between features
- Line plots illustrated trends across continuous variables

Key Findings:

- Both datasets showed natural biological variation in measurements
 - Sex-based differences were evident in both species
 - Age distributions followed expected biological patterns
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5. Machine Learning Models

5.1 Linear Regression Implementation

Model Configuration:

- Algorithm: Ordinary Least Squares Linear Regression
- Train-test split: 80-20 ratio with random_state=42
- Features: All numerical variables except target
- Target: Continuous age values (Rings for Abalone, Age for Crab)

Abalone Results:

- Mean Absolute Error (MAE): [Value from execution]
- Mean Squared Error (MSE): [Value from execution]
- Root Mean Squared Error (RMSE): [Value from execution]
- R^2 Score: [Value from execution]

Crab Results:

- Mean Absolute Error (MAE): [Value from execution]
- Mean Squared Error (MSE): [Value from execution]
- Root Mean Squared Error (RMSE): [Value from execution]
- R^2 Score: [Value from execution]

5.2 Logistic Regression Implementation

Model Configuration:

- Algorithm: Logistic Regression with default parameters
- Binary Classification: Young (below median) vs Old (above median)
- Age threshold determined by median age for each dataset
- Maximum iterations: 1000 to ensure convergence

Binary Classification Setup:

- Abalone: Binary age groups created using median rings value
- Crab: Binary age groups created using median age value
- Target variable: 0 = Young, 1 = Old

Abalone Classification Results:

- Accuracy: [Value from execution]
- Precision: [Value from execution]
- Recall (Sensitivity): [Value from execution]
- F1-Score: [Value from execution]

Crab Classification Results:

- Accuracy: [Value from execution]
- Precision: [Value from execution]
- Recall (Sensitivity): [Value from execution]
- F1-Score: [Value from execution]

5.3 Model Evaluation

Performance Metrics:

- Confusion matrices generated for classification tasks
- Residual analysis performed for regression models
- Cross-validation considerations discussed
- Model comparison visualizations created

Validation Approach:

- Consistent random state (42) for reproducible results
 - Standard 80-20 train-test split across all models
 - Multiple evaluation metrics for comprehensive assessment
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6. Results and Comparison

6.1 Dataset Comparison

Data Quality:

- Both datasets required minimal preprocessing
- Missing value patterns differed between species
- Categorical variable distributions showed species-specific characteristics

Feature Relationships:

- Morphological measurements showed expected biological correlations
- Age-feature relationships varied between species
- Sex-based differences observed in both datasets

6.2 Model Performance Comparison

Linear Regression Performance:

- Comparative R^2 scores indicated model effectiveness for continuous prediction
- RMSE values provided insight into prediction accuracy
- Residual analysis revealed model assumptions validity

Logistic Regression Performance:

- Classification accuracy demonstrated binary age group prediction capability
- Precision-recall metrics showed balanced performance
- Confusion matrices revealed classification patterns

6.3 Best Performing Models

Based on comprehensive evaluation:

- **Linear Regression:** [Dataset] achieved superior performance with $R^2 = [\text{Value}]$
- **Logistic Regression:** [Dataset] demonstrated better classification with Accuracy = [Value]

6.4 Biological Insights

- Physical measurements serve as reliable age indicators in both species
 - Species-specific variation in age prediction accuracy observed
 - Sex-based differences contribute to model performance variations
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7. Conclusion

This comparative analysis successfully demonstrated the application of machine learning techniques to marine biology age prediction problems. Both Linear and Logistic Regression models showed promising results, with species-specific performance variations highlighting the importance of dataset characteristics in model selection.

Key Achievements:

- Successful implementation of complete data science pipeline
- Effective handling of real-world data preprocessing challenges
- Comparative analysis providing insights into marine biology applications
- Robust model evaluation using multiple performance metrics

Research Implications:

- Machine learning approaches show potential for automated age determination in marine species
- Feature selection and engineering critical for optimal performance

- Biological domain knowledge essential for meaningful model interpretation

Future Directions:

- Advanced ensemble methods could improve prediction accuracy
- Feature engineering based on biological knowledge could enhance models
- Cross-species model transfer learning represents an interesting research avenue

The project successfully met all requirements while providing valuable insights into the intersection of machine learning and marine biology research. The systematic approach and comprehensive evaluation demonstrate the potential for data science applications in biological research domains.

Appendices:

- All code implementations following provided patterns
- Complete visualization gallery
- Detailed performance metrics tables
- Statistical analysis results

Note: This report provides the framework structure. Actual numerical results should be filled in after running the Jupyter notebook with your specific datasets.